Water Research 274 (2025) 123135

Contents lists available at ScienceDirect

Water Research

journal homepage: www.elsevier.com/locate/watres

A low-cost soft sensor for sewer flow monitoring — Learning from water level measurements in manholes

Ruozhou Lin [a], Ruihong Qiu [b], Lihan Hu [c], Yaxin Ding [a], Zhiguo Yuan [a,d,*]

[a] School of Energy and Environment, City University of Hong Kong, Hong Kong SAR, China
[b] School of Electrical Engineering and Computer Science, The University of Queensland, St. Lucia, Brisbane, 4072, QLD, Australia
[c] College of Water Conservancy and Hydropower Engineering, Hohai University, Nanjing, 210098, China
[d] State Key Laboratory of Marine Pollution, City University of Hong Kong, Hong Kong SAR, China

ARTICLE INFO

Keywords:
Sewer system
Soft sensor
Flow rate monitoring
Machine learning
Computational fluid dynamics
Noise sensitivity

ABSTRACT

Flow meters are commonly used in manholes to monitor the flow rate for sewer operation and management. However, the large-scale deployment of flow meters in a sewer system is cost-prohibitive due to their high costs and the need for frequent maintenance. This paper proposes a soft sensor that estimates flow rates based on water level measurements in a manhole. The soft sensor is powered by a Multilayer Perceptron (MLP) model, of which the input is a time series of measured water levels, and the output is the corresponding flow rate. The model was trained using flow and water level data collected from three real-life manholes as well as data generated for a simulated manhole using a three-dimensional Computational Fluid Dynamics model. In all cases, the trained MLP produced satisfactory estimations of the flow data in the test phase. Further studies of the simulated manhole showed that the soft sensor is robust against noise and bias associated with the water level and flow measurements. The proposed soft sensor provides a potentially cheaper and more durable alternative to traditional flow meters for sewer applications.

1. Introduction

Sewer systems are among the most critical urban infrastructures, which collect and transport stormwater and wastewater to natural water bodies and wastewater treatment plants, respectively (Butler et al., 2018). Malfunctions of sewer systems could cause urban flooding and/or sewer overflow to the receiving water environment, resulting in property loss, environmental damage, and even casualties (Fu et al., 2008; Lin et al., 2020).

Flow monitoring is critical for sewer operation and management (Butler et al., 2018; Zhang et al., 2021), with ample applications to early warning (Butler et al., 2018), hydrological and hydraulic modelling (Li et al., 2019), risk assessment (Lin et al., 2022), rehabilitation (Lin et al., 2020), and real-time control (Li et al., 2020; Mullapudi et al., 2020), to name a few examples. Flow rates are often measured at manholes with flow meters, especially ultrasonic Doppler flow meters (Bartos & Kerkez, 2021; Bonakdari & Zinatizadeh, 2011). Several reasons hinder the large-scale deployment of flow meters. In addition to their high costs, flow meters typically require frequent maintenance and have low durability, due to the direct contact with wastewater and/or stormwater causing fouling and even failure (Tomperi et al., 2023).

A data-driven soft sensor is a tool that estimates a variable difficult or expensive to measure based on other more easily measurable variables (Fortuna et al., 2007). A water level sensor often costs only a small fraction of a flow meter and is much more durable (Ge et al., 2024; Tomperi et al., 2023). A water level sensor could continuously work for multiple years without noticeable performance degradation, while for flow meters significant drifts can be observed even in the first month (ISO, 2022; Li et al., 2021; Sun et al., 2021). This could also be observed in data used in this study (Section 2.2). Therefore, a soft flow sensor based on water level measurements, and a potentially quantifiable correlation between water levels in a manhole and the flow rate through the manhole, would be attractive. Indeed, water levels have been used to estimate overflow rates at Combined Sewer Overflow (CSO) structures, which divert excessive sewer flow to watercourses (Butler et al., 2018). Kouyi et al. (2005), Fach et al. (2009), and Isel et al. (2013) utilized water level data to estimate flow rate by establishing empirical equations, such as polynomial equations, weir equations, or rating curves.

* Corresponding author.
E-mail addresses: ruozhou.lin@my.cityu.edu.hk (R. Lin), r.qiu@uq.edu.au (R. Qiu), hulihan@hhu.edu.cn (L. Hu), yaxinding3-c@my.cityu.edu.hk (Y. Ding), zhigyuan@cityu.edu.hk (Z. Yuan).

https://doi.org/10.1016/j.watres.2025.123135
Received 31 May 2024; Received in revised form 17 December 2024; Accepted 10 January 2025
Available online 11 January 2025

These soft sensors were calibrated and tested with data generated using Computational Fluid Dynamics (CFD) simulations, displaying acceptable errors typically within 10 %. Subsequently, Ahm et al. (2016) developed a soft sensor for a real-life CSO in Aarhus, Denmark, and validated its effectiveness using data collected from ultrasonic water level sensors and an electromagnetic flow meter. The soft sensor was based on empirical equations calibrated using data generated with a CFD model for this CSO and achieved estimations with errors <3 % for most events when tested with real data.

It should be noted that the abovementioned approaches, designed for CSO structures, are not directly applicable to sewer manholes because flow regimes in manholes is much more complex than in CSO structures. More specifically, the outflow hydraulic conditions in a CSO structure are often free surface flow at a critical state. This ensures that each water depth is paired with exactly one flow rate and vice versa. In other words, the relation is one-to-one or bijective (Chaudhry, 2008). In contrast, the flows into and out of a manhole could be free surface or pressurized flow, at subcritical, critical, or supercritical state, causing a non-bijective and even multivalued relationship between flow rate and water depth (Gironás et al., 2010), which means a water depth could correspond to two or more flow rates. Similar phenomena were observed in river discharge estimation, which were called loop stage-discharge relationships (Braca, 2008). This poses challenges for soft flow sensor development. More rigorous definitions of bijective, non-bijective, and multivalued relationships can be found in Supplementary Information.

Tomperi et al. (2023) attempted to develop a soft flow sensor for a real-life manhole in Finland, where a Doppler device was used to manually measure water depth and flow rate. Linear regression was used to estimate the flow rate from water depth data, achieving a good agreement with a small dataset of 60 data points. While indicating possibilities to develop a soft flow sensor based on water level measurements, the approach is applicable to scenarios where a bijective, linear relationship exists between flow rates and water levels, which is not always satisfied in practice, particularly when manholes are surcharged in wet weather conditions.

This study aims to develop a soft sensor able to estimate manhole flow rates based on a time series of water depths instead of a single water depth. This approach enables a non-bijective and even multivalued relationship to be learned. A common nonlinear machine learning method, namely a Multilayer Perceptron (MLP) model, is adopted. The effectiveness of the soft sensor is demonstrated using three real-life case studies and is optimized through 3D CFD simulation studies.

2. Materials and methods

2.1. General methodology and the machine learning model

In a sewer system, a water level sensor is typically mounted at the upper part of a manhole, while a flow meter is positioned in the downstream pipe, as illustrated in Fig. 1. We aim to develop a soft flow sensor that estimates sewer flow rates using a time series of water levels as the inputs:

$$Q_{i-n} = f(H_{i-m}, \dots, H_{i-2}, H_{i-1}, H_i) \tag{1}$$

where Q_{i-n} $(\mathrm{m}^3/\mathrm{s})$ is the estimated flow rate out of a manhole at time $i-n$; H_i (m) is the water depth in the manhole at time i; f is the function that maps the time series of water depths with a length of $m+1$ to the flow rate; n is a possible delay in the estimation, with $n \le m$.

From a physical perspective, the flow rate Q_{i-n} at time $i-n$ is influenced by water depths in the past steps, i.e., domain of dependence for the Navier–Stokes equations, and will affect water depths in the future steps, namely domain of influence for the Navier–Stokes equations. In other words, both the past and future water depth values contain critical information of the current flow rate. Therefore, the non-bijectivity and multivalued issue identified above may be addressed by incorporating these multiple water depths as soft sensor input. The proposed soft sensor estimates flow rate Q_{i-n} at time $i-n$ based on earlier water level measurements (H_{i-m}, …, H_{i-n-2}, H_{i-n-1}), current measurement (H_{i-n}), and later measurements (H_{i-n+1}, H_{i-n+2}, …, H_{i-1}, H_i), as depicted in Fig. 1. The values of n and m will be decided in model training but are expected to be relatively small as only the water level measurements near $i-n$ have significant influences. The effectiveness of this setting will be demonstrated and validated in the Results section.

A Multilayer Perceptron (MLP) model is employed to learn the relationship between water levels ($H_{i-m}, \dots, H_{i-2}, H_{i-1}, H_i$) and the flow rate ($Q_{i-n}$), as shown in Fig. 1. MLP is one of the most classic and widely used artificial neural networks that consist of at least three layers of neurons: an input layer, a hidden layer, and an output layer (Alpaydin, 2020). The output of the ith layer ($\boldsymbol{z}_i$) are calculated by feeding the output of the previous layer ($\boldsymbol{z}_{i-1}$) into a predefined activation function (σ) with certain weight ($\boldsymbol{W}_{i-1,i}$) and bias ($b_{i-1,i}$):

$$\boldsymbol{z}_i = \sigma\left(\boldsymbol{W}_{i-1,i}\boldsymbol{z}_{i-1} + b_{i-1,i}\right) \tag{2}$$

where $\boldsymbol{W}_{i-1,i}$ and $b_{i-1,i}$ are tunable parameters determined in the training phase. The performance of an MLP model is evaluated by quantifying the discrepancy between model estimation and observed values using loss function, e.g., Mean Squared Error (MSE) and Cross-Entropy (CE).

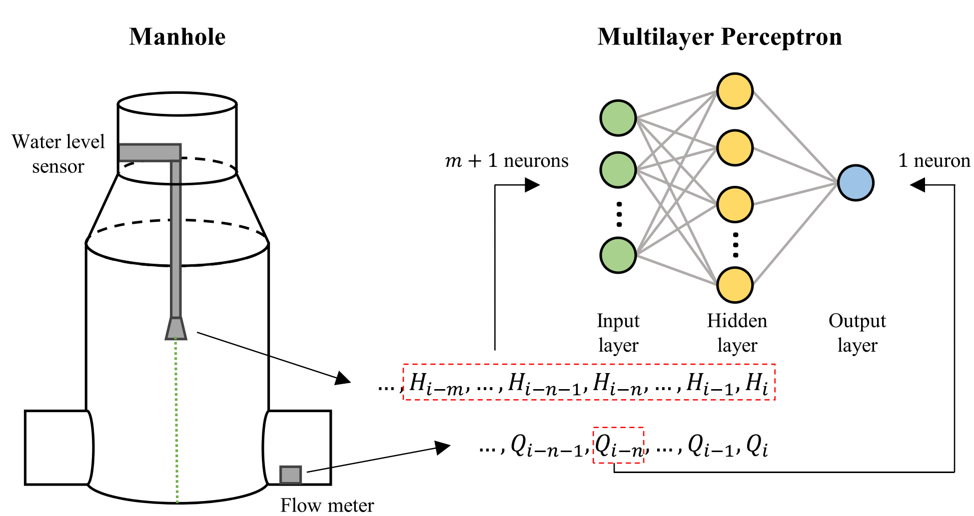

Fig. 1. Schematic of the proposed soft sensor that estimates flow rate based on a time series of measured water levels supported by a multilayer perceptron model.

To minimize the model loss, parameters ($\boldsymbol{W}_{i-1,i}$ and $b_{i-1,i}$) are tuned by a specific backpropagation algorithm, such as the Adam (Kingma & Ba, 2014) and the L-BFGS-B (Liu & Nocedal, 1989) algorithms. This process is also referred to as model training. Generally, MLP is simple yet effective at finding hidden relationships in data (Bisong, 2019).

As depicted in Fig. 1, the input layer reads the time series of the water depths, hence the number of neurons in the input layer equals the number of data points considered (i.e., $m+1$). The output layer contains only one neuron representing the estimated flow rate (Q_{i-n}). Therefore, the model performance (i.e. goodness-of-fit of the model) is mainly determined by two key parameters: size of the input window and structure of the hidden layer(s). Intuitively, a larger window incorporating more measurements provides more information and could consequently improve closeness-of-fit. However, a larger window results in a neural network with more parameters to be trained, which is more prone to overfitting and requires more data and larger computational resources for training. These may cause unsatisfactory model performance. It is therefore critical to choose a proper window size.

In addition to window size, the structure of the hidden layer(s) is also critical. The number of hidden layers should be chosen first. Our strategy was to start with one hidden layer and increase the number of hidden layers if it is incapable of achieving satisfactory performance. Similarly, a proper number of neurons in the hidden layer should be chosen based on model performance. On the one hand, more neurons in the hidden layer theoretically bring better performance; on the other hand, it could cause overfitting and low computational efficiency in practice.

The key MLP parameters, namely the input size ($m+1$), the output delay (n), and the structure of the hidden layer(s) were determined in training and optimization. The layer parameters were initialized using the widely used Kaiming Initialization (He et al., 2015). The activation function was ReLU (Winkler & Le, 2017). The Adam optimizer (Kingma & Ba, 2014) was used with learning rate equaling 10^{-4}. The loss function was Mean Squared Error (MSE). The training batch size was set to 1024. In addition, the Mean Relative Error (MRE) and the Nash-Sutcliffe model efficiency coefficient (NSE) were used for evaluation. We refer to SI for the calculations of MSE, MRE, and NSE. The MLP was implemented using PyTorch, a commonly used AI framework (Paszke et al., 2019). The model was developed following the best practices suggested by Zhu et al. (2023).

2.2. Field data

Water depth and flow rate data are required to train, validate, and test the MLP model. Three manholes (M561, M536, and M570) having significantly different hydraulic characteristics located in three urbanized areas in China were selected. All these manholes have an integrated flow meter (model THWater TWQ from Tsinghuan Smart Water Tech Co., Ltd.) installed to continuously measure both water levels and flow rates for the entire year of 2021 (Fig. S2 in SI) recorded at an interval of 1 min. The sensor measures water levels and flow rates based on the principle of static pressure and Doppler effect, respectively. The data includes both free surface and pressurized flows, and both subcritical and supercritical flows. Although the downstream condition of each manhole is unknown, its effect on the relationship between the water depth and flow rate at the studied manhole is reflected in the data collected in each manhole. Hence, a soft sensor developed based on the data has indirectly considered such influences.

The data from each manhole was first cleaned (see details in SI) and then divided into the training set, validation set, and test set, as will be further described below. Each selected test set was a complete wet event instead of random shuffled and time-discontinuous samples. Subsequently, each training set was normalized individually using Min-max normalization (Alpaydin, 2020).

Manhole M561. Manhole M561 is 5.88 m deep and connected to a circular outflow conduit with a diameter of 1200 mm. In dry weather, the average flow rate is around $4200 \mathrm{m}^3/\mathrm{d}$ with a water depth around 0.15 m. The wet weather flow rate could be 30 times higher with water depth up to 1.75 m, causing severe manhole surcharges. The flow rate data showed noticeable drifts after June 21, 2021, with reasons unknown, hence only records before June 21 were considered to develop the proposed soft sensor. Only a small amount of dry weather flow events was selected for the training set since too many repetitive dry weather events could reduce sensor performance in wet weather, which is typically more important for sewer operation and management. The data series with a total duration of 672 h (i.e., 40,320 data points) were selected and randomly divided into the training set (80 %) and the validation set (20 %) to develop the proposed soft sensor. The trained MLP model was further tested with another wet weather event lasting 96 h (5760 data points).

Manhole M536. Manhole M536 is 5.45 m deep and receives flows from an upstream pump station, which is operated in an intermittent manner with a period around 40 min. The average dry weather flow is approximately $6000 \mathrm{m}^3/\mathrm{d}$, discharged through a 600 mm circular conduit. Since the flow rate data started to drift on Jan 9, 2021, and became observably distorted on Feb 26, 2021, with reasons unknown, data from Jan 1 to Jan 7 were selected and randomly split for training (80 %) and validation (20 %). Data points on Jan 8 were used as the test set. The training, validation, and test set contained 7561, 1891, and 1440 data points, respectively.

Manhole M570. Manhole M570 has a depth of 5.57 m and a circular outflow conduit with a diameter of 600 mm. The average flow rate is around $3500 \mathrm{m}^3/\mathrm{d}$ in dry weather and up to 8 times higher in wet weather. The water depth reached 2.3 m in wet weather, resulting in severe surcharges. Measurements with a total duration of 216 h, covering two wet weather events, were used in model development (i.e., 129,600 data points). The data set was randomly divided into the training set (80 %) and the validation set (20 %). Another 48 h wet weather event was selected as the test set (2880 data points). Although all selected measurements are before March 8, 2021, drifts in flow rate measurements can be observed from Feb 11 onwards with unknown reasons, which may lead to poorer performance of the proposed soft sensor.

2.3. Data generation using a CFD model

The proposed method was also applied to a virtual manhole simulated using 3D Computational Fluid Dynamics (CFD), where the simulated data is noise-free and has much higher measurement frequency (i.e., once per second) compared to Manholes M561, M536, and M570. This case is included to assess the impact of measurement errors, data frequency, and to optimize model performance by adjusting meta parameters.

The first step was to create geometry. The key components of the geometry comprised one upstream conduit, one manhole, and one downstream conduit, as shown in Fig. S1(A). Besides, auxiliary components were used to better apply boundary conditions (Weller et al., 1998). The manhole was set to be 1.6 m wide and 5.0 m high, and both upstream and downstream conduits were 50.0 m long and 0.6 m wide, with slopes equal to 0.005. Other parameters (shown in Fig. S1(A)) were determined in accordance with design manuals (DSD, 2013; 2018) and engineering experiences. The geometry was created using SALOME 9.9.0, a software for geometry creation and mesh generation, among others (Ribes & Caremoli, 2007). As shown in Fig. S1(B), the geometry was then meshed into finite cells using SALOME 9.9.0 with $\Delta x = 10$ cm, which was determined by scale sensitivity analysis (see Fig. S3). Flow rates were synthesized based on real-life flow rates covering both dry and wet weather, resulting in 88 events with a total length of 783.51 h. The flow rate data was then used as boundary conditions in CFD simulations. Other boundary condition settings are listed in Table S1.

In order to simulate flows inside the manhole in three dimensions, Reynolds-averaged Navier-Stokes (RANS) equations were selected as the governing equations, with the standard $k - \epsilon$ model to consider turbulence effects (Beg et al., 2019). In addition, the Volume of Fluid (VOF) method was used to capture the free surface of water flows (Beg et al., 2020). An open source CFD tool OpenFOAM v9 was utilized to numerically solve these equations. More specifically, the governing equations were solved by the solver *interFoam* with detailed settings listed in Table S1, as suggested by previous studies (Beg et al., 2020; 2018; 2019). The time-marching procedure was controlled by a variable time step determined by the maximum Courant-Friedrichs-Lewy (CFL) number, which was set as 0.9. The simulation accuracy was ensured by setting limits for residuals, which are 1.0×10^{-8}, 5.0×10^{-9}, 1.0×10^{-8}, 1.0×10^{-8}, and 1.0×10^{-8} for velocity (u), pressure (p), phase fraction (α), turbulent kinetic energy (k), and turbulent kinetic energy (ϵ), respectively. Simulations were run in parallel on the cluster computing server at the School of Energy and Environment of City University of Hong Kong. During CFD simulations, flow rates and water depths were recorded every second at the manhole outlet and manhole center, respectively, with a total of 2820,636 data points recorded. This dataset consisted of 88 events, divided into 38 training, 10 validation and 40 testing subsets. The large test set ensures the reliability of the proposed soft sensor.

The data generated by CFD simulations had no measurement errors. In contrast, practical measurements often contain noises, biases, and drifting, which may influence the reliability of the proposed soft sensor. Artificial noises were added to data generated by CFD simulations to evaluate the robustness of the soft sensor. According to specifications of commonly used commercial products, the water level sensor errors typically follow the Gaussian distribution with no bias, with a 95 % confidence interval of 3 mm, i.e., with mean (μ) and standard deviation (σ) equalling 0 and 1.5 mm, respectively. On the contrary, the noises associated with a flow meter are often quantified as relative errors. The corresponding Gaussian noises had a 95 % confidence interval of 2 %, namely μ and σ being 0 and 1 %, respectively. In addition, biased errors of -30 mm and -0.08 m^3/s, respectively, were added to the water level and flow rate measurements, according to observable errors in the real-life data.

3. Results

The proposed method was first demonstrated with a 4 min input window with a 2 min estimation delay (2 records from the past steps, 1 current record, and 2 records in the future steps, i.e., 5 input neurons) and a hidden layer containing 128 neurons. This is a near-optimal setting according to the results to be shown in Section 3.4.

3.1. Nonlinear and multivalued relationship between flow rates and water depths

Fig. 2 depicts relationships between flow rates and water levels for Manholes M561, M536, M570, and the simulated manhole. The area with high scatter densities in the left part of each graph mainly represents dry weather events that are typically more frequent and have lower flow rates and water depths. On the contrary, the region with a sparse scatter density in the right part is composed of wet weather events where flow rates and water depths were much higher. The nonlinear and multivalued relationships are more pronounced in periods with higher flows. This indicates that the linear regression method (Tomperi et al.,

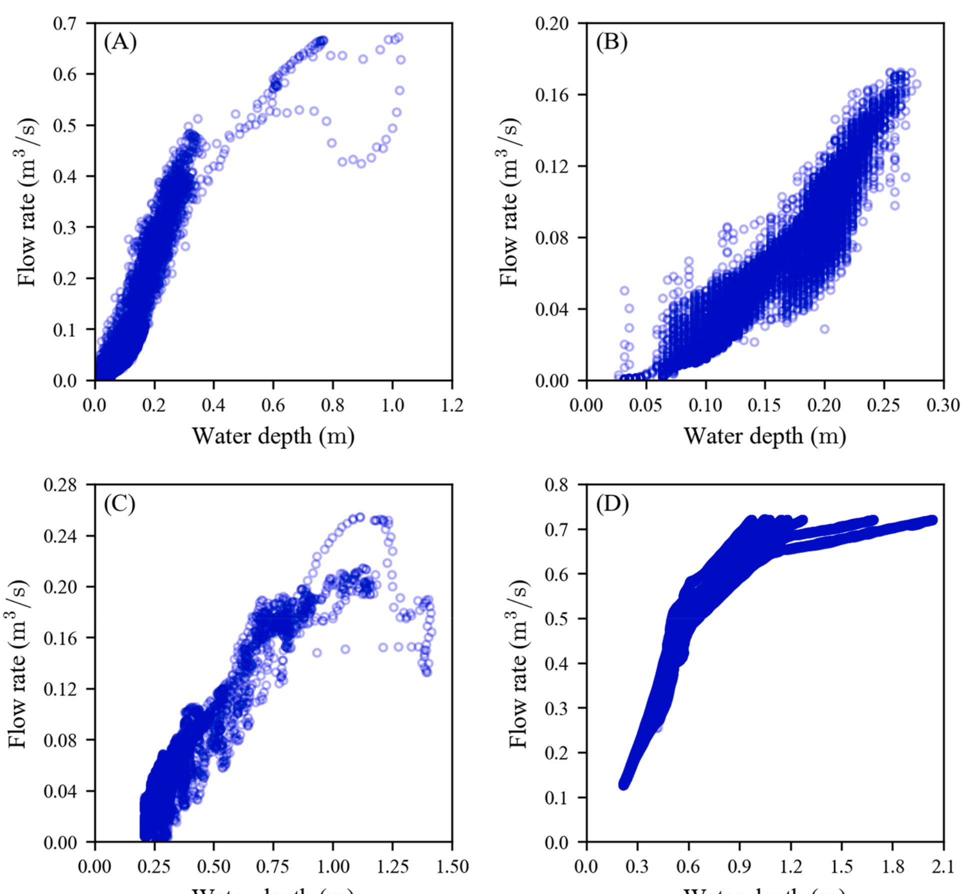

Fig. 2. Datasets for the development of the proposed soft sensors for Manhole M561 (A), M536 (B), M570 (C), and the simulated manhole (D). Strong nonlinear and multivalued relationships are observed.

2023) may be insufficient to estimate a flow rate with an acceptable accuracy based on a single water depth data point, this will be further discussed in the next section.

3.2. Flow rates estimation using MLP

Manhole M561. Fig. 3(A) depicts the estimated flow rate by the soft sensor in comparison to the measured flow rate for Manhole M561. A good agreement between the estimated and measured flow rates can be observed, without significant under- or over-estimation ($R^2 = 0.9791$), indicating satisfactory performance of the proposed soft sensor. As shown in Fig. 3(B), the proposed soft sensor was applied to the test set to further verify its effectiveness. The flow rate series "measured" by the proposed soft sensor agreed well with the actual measurements for both dry and wet weather flows, with NSE and MRE values of 0.9788 and 7.85 %, respectively (see Table 1). This implies that the proposed soft sensor was effective and reliable for Manhole M561. In addition, the estimated flow rates closely matched the measured data points for both dry and wet weather flows with an MRE of 13.57 % (Fig. 3(C)), indicating that the proposed soft sensor could deal with the nonlinear and multivalued properties of complex flows during application.

Manhole M536. As shown in Fig. 4(A), the flow rates estimated by the soft sensor for M536 were also in a good agreement with the measured flow rates ($R^2 = 0.9712$). Different from M561, flows in manhole M536 were affected by a pump upstream, resulting in regular sharp changes, as shown in Fig. 4(B). The proposed soft sensor was capable of dealing with this complex scenario, achieving NSE and MRE values of 0.9610 and 9.81 %, respectively. The soft sensor successfully captured the nonlinear and multivalued flow patterns with MRE equaling 9.02 % (Fig. 4(C)).

Manhole M570. Fig. 5 shows the results of Manhole M570, where the estimated and observed flow rates maintained the same trend, with NSE and MRE equaling 0.8906 and 16.4 %, respectively. As can be seen, the soft sensor performance was slightly lower than the soft sensors for M561 and M536. This could be attributed to the poorer data quality at Manhole M570. Indeed, drifts in flow rate measurements were observed for the test set (see Fig. S2(C1)), where the actual flow rates should be systematically higher. This accounts for the 'overestimated' flow rates in Fig. 5A and 5B and Fig. 5C (dry weather flows and the peak flow around 0.20 m^3/s).

To further illustrate its effectiveness, the proposed soft sensor was applied to Manhole M570 in a period when the flow meter malfunctioned (April 25 to April 29). As depicted in Fig. 5(D), the measurements from the water level sensor showed a significant diurnal pattern, but the measured flow rates were zero most of time, which cannot be possible given the non-zero water depth in this period (data not shown). With the help of the proposed soft sensor, the flow rates in this period could be reconstructed.

Choice of time series. In the above-trained soft sensors, we used a time series comprising five water depths, namely H_{i-2}, H_{i-1} H_i, H_{i+1}, and H_{i+2}, to estimate Q_i, which incurs a 2 min delay. Here, two additional soft sensors were trained, which estimate Q_i based on the 'past' time

Table 1
Performance of soft sensors with different settings.

Criterion		NSE		
Sensor		M561	M536	M570
1	$Q_i = kH_i + b$	0.7216	0.8961	0.7433
2	$Q_i = f(H_i)$	0.9562	0.9107	0.8487
3	$Q_i = f(H_{i-2}, H_{i-1}, H_i)$	0.9739	0.9410	0.8749
4	$Q_i = f(H_i, H_{i+1}, H_{i+2})$	0.9738	0.9285	0.8722
5	$Q_i = f(H_{i-2}, H_{i-1}, H_i, H_{i+1}, H_{i+2})$	0.9788	0.9610	0.8906
6	$Q_i = f(H_{i-4}, H_{i-3}, H_{i-2}, H_{i-1}, H_i)$	0.9747	0.9508	0.8798

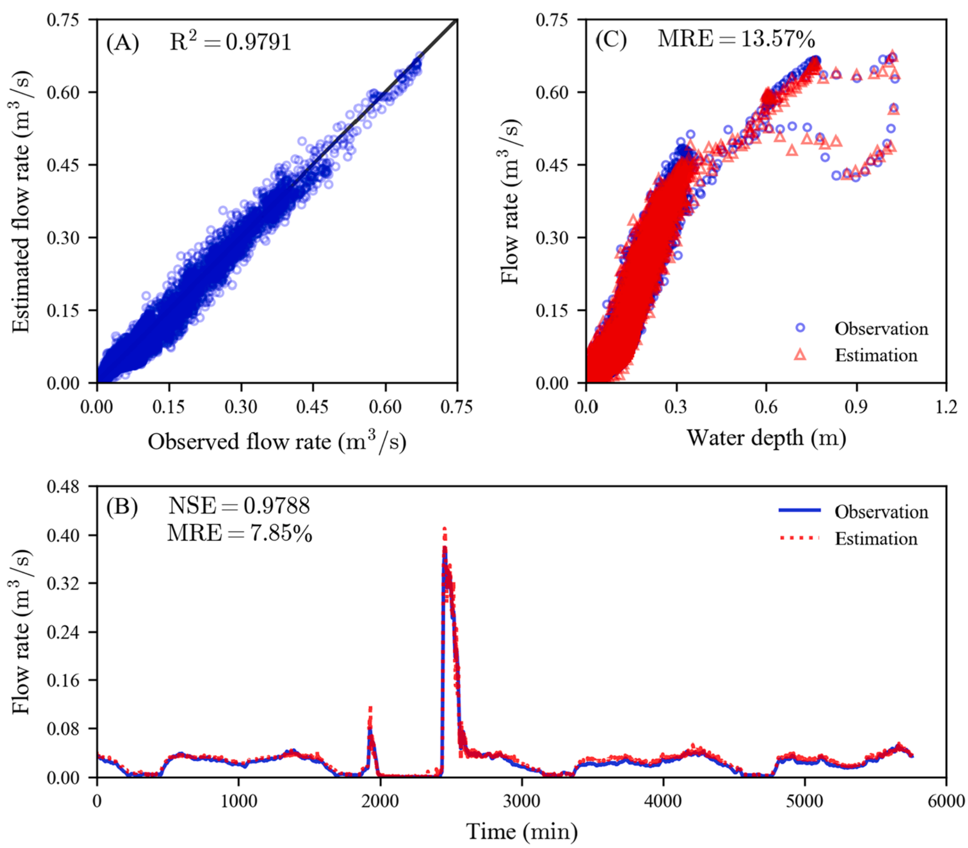

Fig. 3. Performance of the proposed soft flow sensor for Manhole M561 regarding estimation accuracy (A), effectiveness of continuous estimation (B), and nonlinear and multivalued relationship (C). (A) and (C) both display the whole dataset, while (B) only depicts the test set.

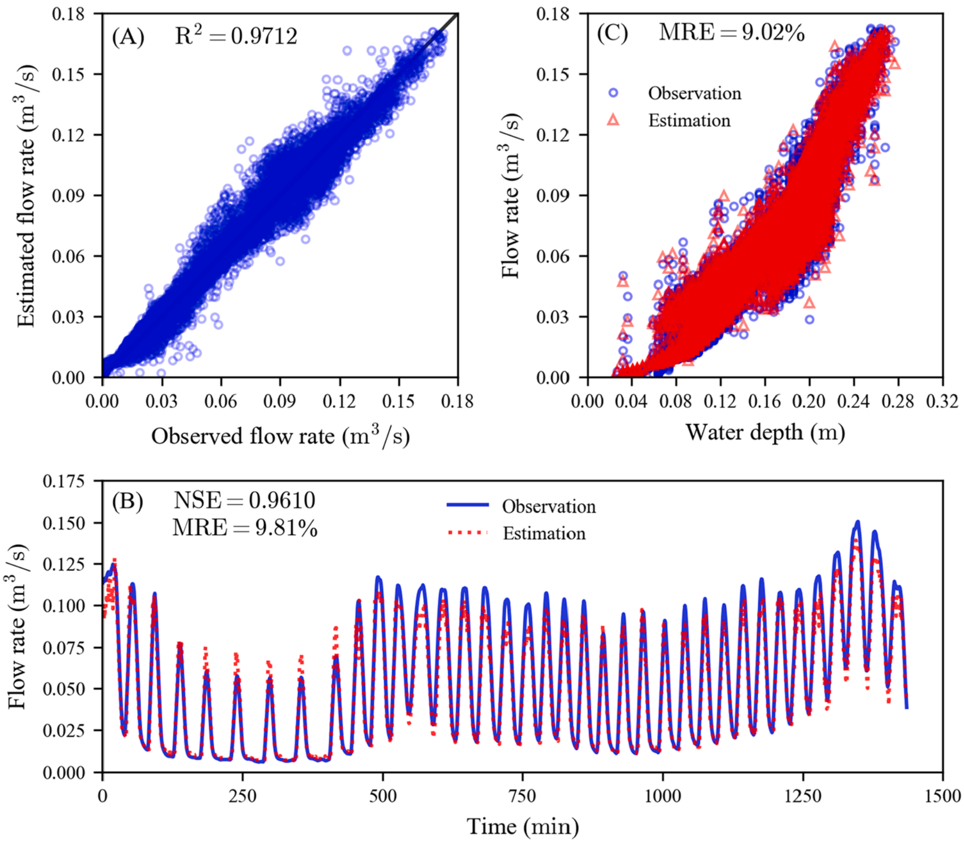

Fig. 4. Performance of the proposed soft flow sensor for Manhole M536 regarding estimation accuracy (A), effectiveness of continuous estimation (B), and nonlinear and multivalued relationship (C). (A) and (C) both display the whole dataset, while (B) only depicts the test set.

series (H_{i-2}, H_{i-1}, and H_i) and 'future' time series (H_i, H_{i+1}, and H_{i+2}), respectively. This is to investigate the relative importance of the 'past' and 'future' water depth information. For comparison, we also trained two soft sensors that estimate Q_i based on a single water depth H_i, with linear and nonlinear (using MLP) regression, respectively. The results are shown in Fig. 6 and Table 1.

Based on a single water depth, both Sensor 1 and Sensor 2 failed to capture the multivalued relationship between the flow rate and the water depth (Fig. 6). Further, Sensor 1 failed to capture the nonlinear relationship between the flow rate and the water depth. In contrast, the remaining three sensors captured the multivalued relationship to various extent by using a continuous sequence of water depths as the input, suggesting the importance of using a time series rather than a single point. Besides, Sensor 5 outperformed Sensor 3 and Sensor 4 by incorporating both 'past' and 'future' water depth information, indicating the necessity of adding an output delay for the soft sensor performance. Take Manhole M536 as an example (Fig. 6(B) and Table 1). Sensor 3 and Sensor 4 achieved reasonable NSE values of 0.9410 and 0.9285, respectively, by addressing the multivalued relationship using backward- and forward-looking time series as the input, respectively. In comparison, Sensor 5 further enhanced sensor performance by incorporating both 'past' and 'future' water depth data as the input, achieving an NSE equaling 0.9610. Although the proposed soft sensor inevitably causes a 2 min delay in flow estimation, it is acceptable for most engineering demands related to sewer operation and management. In case such a delay is not acceptable, Sensor 3 could be used, with a slightly reduced performance.

To confirm the superiority of Sensor 5 was due to its consideration of both past and future information, Sensor 6 with the same number of features but without considering future information was also evaluated, as shown in Table 1. Sensor 6 generally outperformed Sensor 3 and 4 due to the usage of more features. However, Sensor 5 was still superior to Sensor 6, suggesting that it is beneficial to use physical information from both the domain of dependence and the domain of influence.

3.3. Optimizing key meta parameters in the MLP-based soft sensor

The simulated data of the virtual manhole were used to optimize the key parameters in the MLP model, namely the window size, the time step, and the number of neurons in the hidden layer.

Window size. A larger input window, which incorporates more information, may improve soft sensor performance. However, it also increases the difficulty of model training. As shown in Fig. 7(A), the optimal window size is around 4 min. When a window size of 2 min was used, larger errors occurred, likely because some key information was not included in the input. Similarly, a too large window (≥ 12 min) also caused poor performance due to increased number of parameters to be trained. Considering the reduced training effort for a smaller window size, 4 min is recommendable, which is the window size used in Section 3.1.

Sampling frequency. The sensor sampling frequency could also be a critical factor. In the development of soft sensors for the three manholes, we were limited to 1 record per minute, which was the sampling frequency in data collection. With the simulated manhole, we were able to use sampling intervals ranging from 1 s to 2 min. Fig. 7(B) shows that sampling intervals of 10 s to 1 min resulted in similar soft sensor performance. Considering a lower sampling frequency reduces the computational load for MLP training and risks of over-fitting, a sampling frequency of 1 record per minute appears to be a suitable choice for the intended application.

Number of neurons. Fig. 7(C) depicts the sensor performance as a function of the number of neurons (32, 64, 128, 256, 512, 800, and 1024) in the hidden layer of the MLP. The sensor performance improved when the neurons were increased from 32 to 64 to 128, but did not

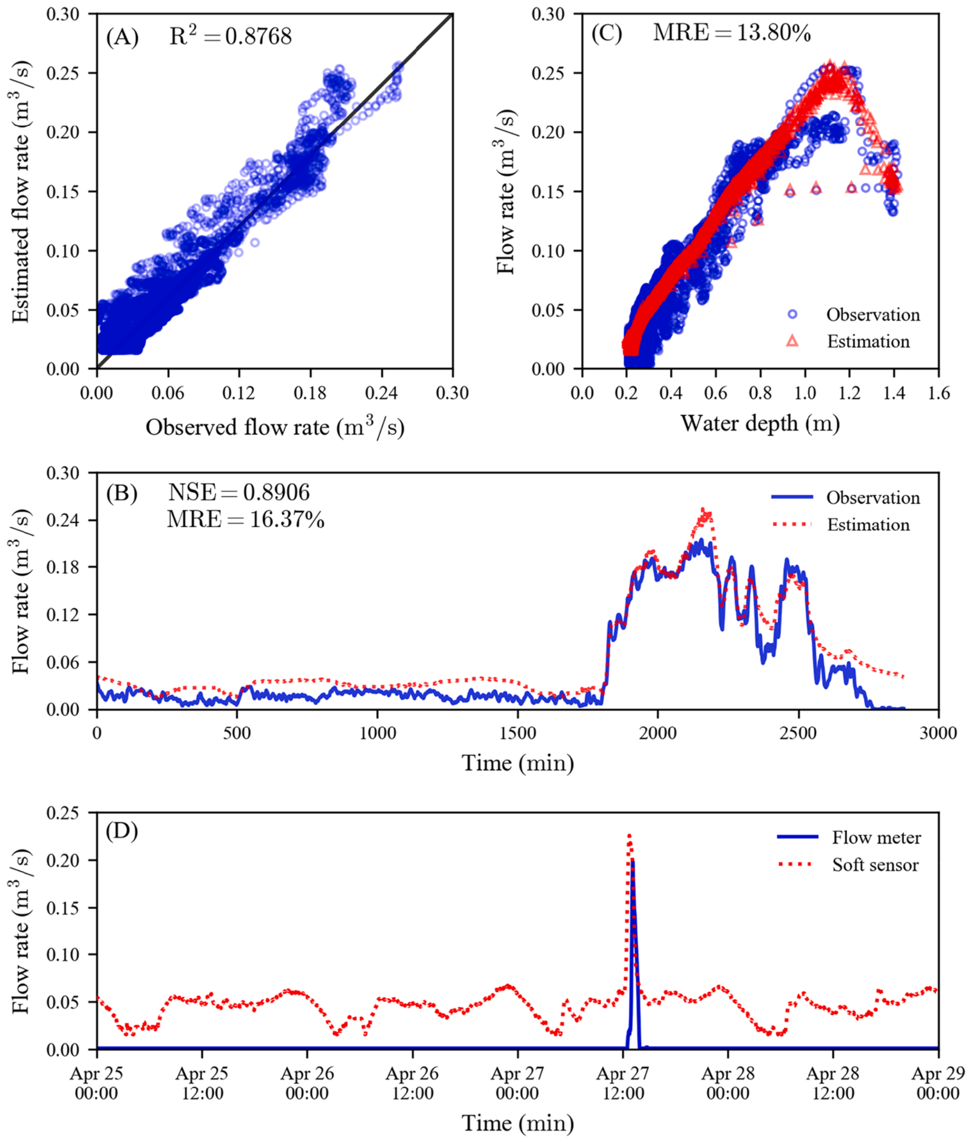

Fig. 5. Performance of the proposed soft flow sensor for Manhole M570 regarding estimation accuracy (A), effectiveness of continuous estimation (B), and nonlinear and multivalued relationship (C). (A) and (C) both display the whole dataset, while (B) only depicts the test set., and flow rates reconstruction for a period when the flow meter was malfunctioning (D). (A) and (C) both display the whole dataset, while (B) only depicts the test set.

further improve when the number of neurons was further increased. Therefore, 128 neurons represent a good balance between the sensor performance and the computational costs, and between the sensor accuracy and risks of over-fitting. Considering the adequate sensor performance, we did not attempt to increase the number of hidden layers.

3.4. Robustness against measurement errors

The simulated manhole was used to analyze sensitivity of the soft sensor to measurement noises and biases. Without measurement errors, the soft sensor was able to accurately estimate flow rates, as shown in Fig. S4. By adding Gaussian noises to the water depth measurements, the MRE slightly increased from 2.88 % to 2.90 %, and the NSE decreased from 0.9372 to 0.9337, in the test cases, indicating that the soft sensor is insensitive to random measurement errors. We further challenged the soft sensor by increasing the 95 % confidence interval from 3 mm to 10 mm, which is larger than all product specifications we have found, the MRE and NSE only increased to 3.26 % and decreased to 0.9223, respectively, indicating the soft sensor still has an acceptable level of accuracy. When the Gaussian noise was added to the flow data used for training, the MRE increased from 2.88 % to 3.32 %, and the NSE decreased from 0.9372 to 0.9318. By simultaneously adding Gaussian noises to both the flow rate and water depth data, MRE and NSE only increased to 3.38 % and decreased to 0.9262, respectively. All these testing results suggest that the soft sensor is insensitive to random measurement noises in both the flow rate and water level measurements. Given that a 1-min sampling frequency is sufficient to develop an effective soft sensor, the robustness of the proposed soft sensor against random noises can be further enhanced by increasing measurement frequency and incorporating a noise reduction filter.

By using a water level sensor with a fixed bias (− 30 mm), the developed soft sensor achieved satisfactory performance with MRE and NSE equaling 3.20 % and 0.9253, respectively. This was likely achieved because the soft sensor was learned from the biased data. We further tested the case where the soft sensor was trained using water depth data without a bias, but tested with water depth data with a bias of − 30 mm. This represented a 9.00 % mean relative error (MRE) in the model input. The estimated flow rates were systematically lower than the actual flow rates with MRE equaling 11.80 %. This implies that the proposed soft sensor propagated but did not substantially amplify the biased error in

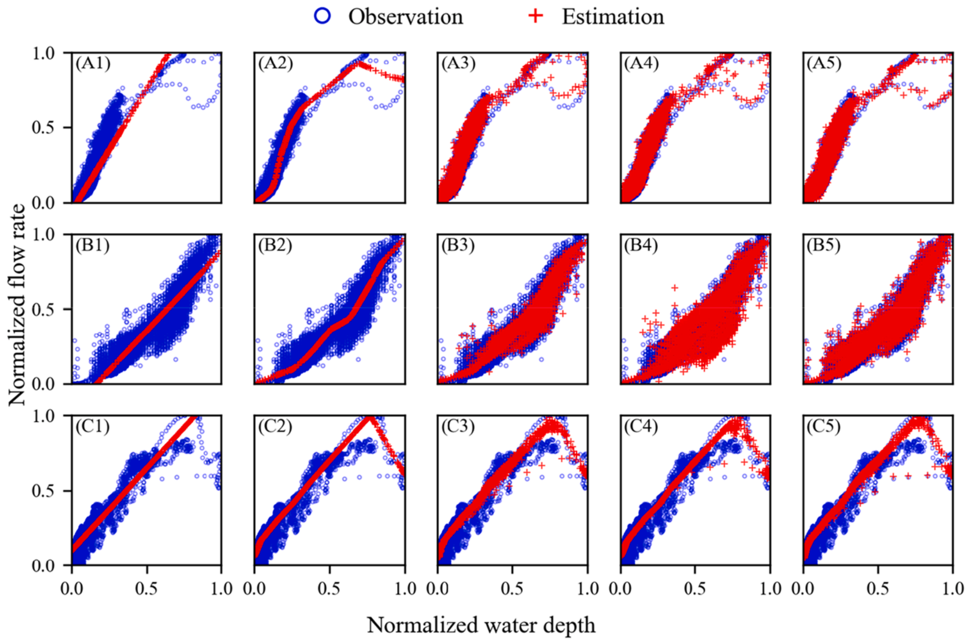

Fig. 6. Performance of different type of soft sensors for Manholes M561 (A), M536 (B), and M570 (C). The index 1, 2, 3, 4, 5 represents soft sensors based on $\boldsymbol{Q_i} = \boldsymbol{kH_i} + \boldsymbol{b}$, $\boldsymbol{Q_i} = \boldsymbol{f}(\boldsymbol{H_i})$, $\boldsymbol{Q_i} = \boldsymbol{f}(\boldsymbol{H_{i-2}}, \boldsymbol{H_{i-1}}, \boldsymbol{H_i})$, $\boldsymbol{Q_i} = \boldsymbol{f}(\boldsymbol{H_i}, \boldsymbol{H_{i+1}}, \boldsymbol{H_{i+2}})$, and $\boldsymbol{Q_i} = \boldsymbol{f}(\boldsymbol{H_{i-2}}, \boldsymbol{H_{i-1}}, \boldsymbol{H_i}, \boldsymbol{H_{i+1}}, \boldsymbol{H_{i+2}})$, respectively.

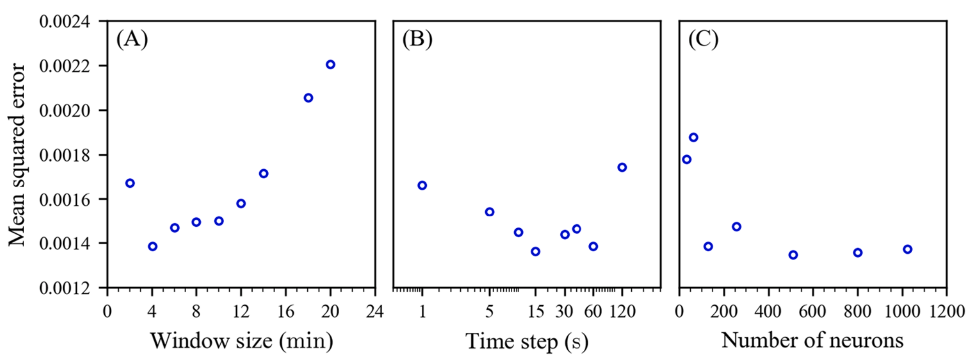

Fig. 7. Performance of the proposed soft senser under different settings. (A) Input window size; (B) Sampling frequency; (C) Number of neurons in the hidden layer.

the water level measurements. We also added a bias of -0.08 m^3/s to the flow data used in the sensor training, which represents a MRE of 26.18 % in the model input. This resulted in an MRE of 37.62 %, indicating the error was amplified by 34.55 %. This means that the quality of the flow rate data is critical for the sensor performance.

4. Discussion

The present study illustrated the feasibility and effectiveness of using a simple soft sensor to estimate flow rates based on water level measurements in manholes. A MLP model that takes a time series of water depths as the input was used for this purpose. The proposed method achieved reliable estimation for three real-life manholes with different hydraulic characteristics under various conditions. The proposed method is robust against measurement errors associated with water level measurements but requires unbiased flow data for the model training.

Intriguingly, the relationship between water depths and flow rates is not only nonlinear but also multivalued (Fig. 2), which makes it challenging to develop an effective soft flow sensor. With our approach, the MLP model was able to deal with the nonlinear behavior (Fig. 6). The use of a time series of water depths as the input, as opposed to a single water depth measurement, enabled the MLP to successfully capture the multivalued relationship (Figs. 3–6). From a physical perspective, flows in a manhole can be described by a set of nonlinear Partial Differential Equations (PDEs), for example the Navier–Stokes equations. As discussed in Section 2, both the past and future water depths carry important information related to the current flow rate via the domain of dependence and the domain of influence. This explains why the use of a time series of water depths comprising both the 'past' and 'future' data as the input could significantly improve the sensor performance (Fig. 6 and Table 1).

The proposed soft sensor is potentially more cost-effective than a flow meter. It is commonly known that a water level sensor costs substantially less than a flow meter (by more than one order of magnitude). Also, a Doppler flow meter requires weekly inspection and maintenance (Li et al., 2021), incurring substantial maintenance costs. In comparison, an ultrasonic water level sensor generally requires annual maintenance (personal communication with industry collaborator). However, the soft sensor requires regular retraining, which contributes to its 'maintenance' costs. The simple MLP structure means that each retraining is straightforward, requiring only hours to complete, so the main costs would be related to the collection of new datasets for retraining, validation and testing. If the retraining is infrequent (e.g. once every few years) to take into consideration changes in climate, demographics, and in the physical infrastructure, the cost for reinstallation of a flow meter for data collection is expected to be much lower than the permanent use of the flow sensor. In contrast, if frequent retraining (e.g. multiple times a year) is needed due to, for example, rapid buildup of sewer sediments, the soft sensor will not be as cost-effective. In fact, it could be a good option to permanently install a flow meter and a water level sensor in parallel. The frequently calibrated soft sensor could be used to verify the flow meter data to achieve better data quality assurance.

For on-site implementation of the soft sensor, the MLP model could be straightforwardly implemented into an existing SCADA system. For sewer sites without a SCADA system, the microprocessor in the water level sensor could be used to run the proposed model with negligible additional costs (basically two matrix linear operations and one nonlinear rectifier operation). As such, the additional "platform as a service" cost (IoT platforms, network services, cyber security, and data storage, to name a few) is avoided, which typically accounts for a significant portion of the total cost of most digital solutions.

In this work, the soft sensor is powered by an MLP, which is able to take a time series as the input. However, an MLP model does not explicitly treat its input as a time series. More advanced machine learning models, such as the Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and the Transformer, are now available, which are designed to learn from time series data. These models could potentially further improve the sensor performance. This should be studied in the future.

Both water level and flow rate data are required for the soft sensor training. This could potentially hinder the widespread application of the method, as a comprehensive measurement campaign is required for each manhole requiring such a soft sensor. Each campaign should last for a period covering dry weather and a variety of wet weather conditions. While such an exercise is more cost effective than permanently deploying a flow sensor (due to the low costs and low maintenance of water level sensors), it nevertheless incurs considerable labor costs. One potential solution is to train the model using data generated from a digital twin of the manhole and the connecting conduits. The hydraulics in a sewer system is well understood, and detailed mechanistic models are available. Indeed, we have implemented 3D models for the simulated manhole and the connecting conduits in this study. This approach could be augmented to build digital twins of real systems based on the real geometry and flow conditions.

Digital twins could also provide a solution for the data sufficiency issue. Although the data for the entire year of 2021 was collected, only a small fraction of the data (less than one month) was considered reliable for each manhole as flow rate data drifted. This is a common problem for sewer flow meters due to their direct contact with wastewater causing low durability. To develop a soft flow sensor that is reliable for long-term applications (e.g., over a decade), a much larger dataset with a high quality might be necessary, which could be difficult to collect using current flow meters. Digital twins, following successful calibration using an initial set of high-quality data, could serve as reliable data sources since sensor drift and noise are avoided.

5. Conclusions

A soft flow sensor for flow rate estimation in manholes was designed and implemented under variable conditions. The following conclusions are drawn:

- An MLP model taking a short time series of water depths (4 min in the studied cases) as the input can effectively estimate the flow out of the manhole, as demonstrated in three real-life case studies with different hydraulic characteristics under various weather conditions.
- The relationship between the water depth and the flow rate in a manhole is nonlinear and multivalued. The soft sensor effectively accounts for the nonlinearity using an MLP model and the multivalued property by taking a time series of water depths as the input.
- The soft sensor performance is enhanced when both the 'future' and 'past' water depth data are used as the input. This only results in a short delay (2 min in the studied cases) in the estimated flow rates.
- The proposed soft sensor is robust against measurement noises and biases associated with water level sensor, and the measurement noises associated with the flow sensor. However, it is relatively sensitive to biased flow measurements, and amplifies the error by 34.55 % in our simulated cases. This means systematic errors associated with the flow meter need to be minimized when collecting the flow data for model training.

CRediT authorship contribution statement

Ruozhou Lin: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Ruihong Qiu:** Writing – review & editing, Methodology, Conceptualization. **Lihan Hu:** Methodology. **Yaxin Ding:** Methodology. **Zhiguo Yuan:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank Beijing Tsinghuan Smart Water Tech Co., Ltd. for providing the field data for the soft sensor development. Zhiguo Yuan is a Global STEM Professor jointly funded by the Innovation, Technology and Industry Bureau ("ITIB") and Education Bureau ("EDB") of the Government of the Hong Kong Special Administrative Region (GSP 266), and acknowledges financial support from the Hong Kong Jockey Club for the JC STEM Lab of Sustainable Urban Water Management.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2025.123135.

Data availability

The authors do not have permission to share data.

References

Ahm, M., Thorndahl, S., Nielsen, J.E., Rasmussen, M.R., 2016. Estimation of combined sewer overflow discharge: a software sensor approach based on local water level measurements. Water Sci. Technol. 74 (11), 2683–2696.
Alpaydin, E., 2020. Introduction to Machine Learning. MIT Press.
Bartos, M., Kerkez, B., 2021. Pipedream: an interactive digital twin model for natural and urban drainage systems. Environ. Model. Softw. 144.
Beg, M., Rubinato, M., Carvalho, R., Shucksmith, J., 2020. CFD modelling of the transport of soluble pollutants from sewer networks to surface flows during urban flood events. Water 12 (9). . (Basel).
Beg, M.N.A., Carvalho, R.F., Leandro, J., 2018. Effect of surcharge on gully-manhole flow. J. Hydro Environ. Res. 19, 224–236.
Beg, M.N.A., Carvalho, R.F., Leandro, J., 2019. Effect of manhole molds and inlet alignment on the hydraulics of circular manhole at changing surcharge. Urban Water J. 16 (1), 33–44.
Bisong, E., 2019. Building Machine Learning and Deep Learning Models On Google cloud Platform. Springer.
Bonakdari, H., Zinatizadeh, A.A., 2011. Influence of position and type of Doppler flow meters on flow-rate measurement in sewers using computational fluid dynamic. Flow Meas. Instrum. 22 (3), 225–234.
Braca G. (2008) Stage-discharge relationships in open channels: practices and problems, Univ. degli Studi di Trento, Dipartimento di Ingegneria Civile e Ambientale ….
Butler, D., Digman, C., Makropoulos, C., Davies, J.W., 2018. Urban Drainage. Crc Press.
Chaudhry, M.H., 2008. Open-channel Flow. Springer.

DSD 2013 Sewerage manual: key planning issues and gravity collection system. Department, D.S. (ed).
DSD 2018 Stormwater drainage manual: planning, design and management. Department, D.S. (ed).
Fach, S., Sitzenfrei, R., Rauch, W., 2009. Determining the spill flow discharge of combined sewer overflows using rating curves based on computational fluid dynamics instead of the standard weir equation. Water Sci. Technol. 60 (12), 3035–3043.
Fortuna, L., Graziani, S., Rizzo, A., Xibilia, M.G., 2007. Soft Sensors for Monitoring and Control of Industrial Processes. Springer.
Fu, G., Butler, D., Khu, S.-T., 2008. Multiple objective optimal control of integrated urban wastewater systems. Environ. Model. Softw. 23 (2), 225–234.
Ge, J., Li, J., Qiu, R., Shi, T., Zhang, C., Huang, Z., Yuan, Z., 2024. A data-driven method for estimating sewer inflow and infiltration based on temperature and conductivity monitoring. Water Res. 261, 122002.
Gironás, J., Roesner, L.A., Rossman, L.A., Davis, J., 2010. A new applications manual for the storm water management model (SWMM). Environ. Model. Softw. 25 (6), 813–814.
He K., Zhang X., Ren S. and Sun J. 2015 Delving deep into rectifiers: surpassing human-level performance on ImageNet classification, pp. 1026–1034.
Isel, S., Dufresne, M., Bardiaux, J.B., Fischer, M., Vazquez, J., 2013. Computational fluid dynamics based assessment of discharge-water depth relationships for combined sewer overflows. Urban Water J. 11 (8), 631–640.
ISO I.O.F.S. 2022 Hydrometry — Water level measuring devices.
Kingma D.P. and Ba J. 2014. Adam: a Method for stochastic optimization. CoRR abs/1412.6980.
Kouyi, G.L., Vazquez, J., Gallin, Y., Rollet, D., Sadowski, A.G., 2005. Use of 3D modelling and several ultrasound sensors to assess overflow rate. Water Sci. Technol. 51 (2), 187–194.
Li, J., Li, W., Chang, X., Sharma, K., Yuan, Z., 2020. Real-time predictive control for chemical distribution in sewer networks using improved elephant herding optimization. IEEe Trans. Ind. Inform. 18 (1), 571–581.
Li, J., Sharma, K., Liu, Y., Jiang, G., Yuan, Z., 2019. Real-time prediction of rain-impacted sewage flow for on-line control of chemical dosing in sewers. Water Res. 149, 311–321.
Li, Z., Jiang, M., Pan, R., Ji, X., Yan, Y., Zhao, D., Li, M., 2021. Monitoring and operating condition analysis of urban sewer network in Yanqing District, Beijing. China. Water Wastewater 37 (20), 99–105.
Lin, R., Zheng, F., Ma, Y., Duan, H.-F., Chu, S., Deng, Z., 2022. Impact of spatial variation and uncertainty of rainfall intensity on urban flooding assessment. Water Resour. Manag. 36 (14), 5655–5673.
Lin, R.Z., Zheng, F.F., Savic, D., Zhang, Q.Z., Fang, X.G., 2020. Improving the effectiveness of multiobjective optimization design of urban drainage systems. Water. Resour. Res. 56 (7).
Liu, D.C., Nocedal, J., 1989. On the limited memory BFGS method for large scale optimization. Math. Program 45 (1), 503–528.
Mullapudi, A., Lewis, M.J., Gruden, C.L., Kerkez, B., 2020. Deep reinforcement learning for the real time control of stormwater systems. Adv. Water Resour. 140.
Paszke A., Gross S., Massa F., Lerer A., Bradbury J., Chanan G., Killeen T., Lin Z., Gimelshein N., Antiga L., Desmaison A., Köpf A., Yang E., DeVito Z., Raison M., Tejani A., Chilamkurthy S., Steiner B., Fang L., Bai J. and Chintala S. 2019 PyTorch: an imperative style, high-performance deep learning library, p. arXiv:1912.01703.
Ribes, A., Caremoli, C., 2007. Salome Platform Component Model For Numerical Simulation. IEEE, pp. 553–564.
Sun, B., Chen, S., Liu, Q., Lu, Y., Zhang, C., Fang, H., 2021. Review of sewage flow measuring instruments. Ain Shams Eng. J. 12 (2), 2089–2098.
Tomperi, J., Rossi, P.M., Ruusunen, M., 2023. Estimation of wastewater flowrate in a gravitational sewer line based on a low-cost distance sensor. Water Pract. Technol. 18 (1), 40–52.
Weller, H.G., Tabor, G., Jasak, H., Fureby, C., 1998. A tensorial approach to computational continuum mechanics using object-oriented techniques. Comput. Phys. 12 (6), 620–631.
Winkler, D.A., Le, T.C., 2017. Performance of deep and shallow neural networks, the universal approximation theorem, activity cliffs, and QSAR. Mol. Inform. 36 (1–2), 1600118.
Zhang, Q., Zheng, F., Jia, Y., Savic, D., Kapelan, Z., 2021. Real-time foul sewer hydraulic modelling driven by water consumption data from water distribution systems. Water Res. 188, 116544.
Zhu, J.J., Yang, M., Ren, Z.J., 2023. Machine learning in environmental research: common pitfalls and best practices. Environ. Sci. Technol. 57 (46), 17671–17689.